PRACTICAL LAB: JAVASCRIPT OPERATORS MASTERY

Course: DFK20013 Web Design Technology

Topic: 3.1.1 Variables, Data Types, and Operators

Duration: 60 – 90 Minutes

Student Name	**KIRTHIKA A/P ARASU**
Registration No	**10DIT25F1176**
Date	**2.3.2026**

---

1. Learning Outcomes

By the end of this lab, students should be able to:

- Implement **Arithmetic Operators** for calculations.
- Use **Assignment Operators** to update values efficiently.
- Control application states using **Increment/Decrement**.
- Validate user access using **Relational** and **Logical** operators.

---

2. Practical Task: The All-in-One Operator Workshop

Create a file named lab_operators.html. Copy and complete the coding below.

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <title>DFK20013: JS Operators Workshop</title>

    <style>

        body { font-family: 'Segoe UI', sans-serif; max-width: 850px;
margin: 2rem auto; background-color: #f0f2f5; padding: 20px; color: #333;
}

        section { background: white; padding: 25px; border-radius: 12px;
margin-bottom: 25px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }

        h2 { color: #2c3e50; border-left: 5px solid #3498db; padding-
left: 10px; margin-top: 0; }

        .output-box { background: #2c3e50; color: #27ae60; padding: 15px;
border-radius: 6px; font-family: 'Courier New', monospace; margin-top:
10px; }
```

```
        .short-note { background: #fff3cd; color: #856404; padding: 15px;
border-radius: 6px; margin-top: 15px; border-left: 5px solid #ffc107;
display: none; font-size: 0.9rem; }

        input { padding: 8px; width: 100px; border-radius: 4px; border:
1px solid #ccc; }

        button { padding: 8px 20px; background: #3498db; color: white;
border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }

        button:hover { background: #2980b9; }

        code { background: #eee; padding: 2px 4px; border-radius: 3px;
font-weight: bold; color: #d63384; }

    </style>

</head>
```

```
<body>



    <header style="text-align: center; margin-bottom: 30px;">

        <h1>JavaScript Operators Mastery</h1>

        <p>Follow the steps and observe the Console and UI outputs.</p>

    </header>
```

```
    <section>

        <h2>Task 1: Basic Arithmetic (Check Console)</h2>

        <p>This task demonstrates background calculations using
<code>+</code>, <code>-</code>, <code>*</code>, <code>/</code>,
<code>**</code>, and <code>%</code>.</p>

        <button onclick="runTask1()">Run Console Logic</button>

    </section>
```

```
    <section>

        <h2>Task 2: POS System (Arithmetic & Assignment)</h2>

        <p>Price: RM 50.00. <strong>Promo:</strong> Buy > 4 units to get
RM 20.00 discount.</p>

        Quantity: <input type="number" id="qty" value="1">

        <button onclick="runTask2()">Calculate Total</button>

        <div id="outPOS" class="output-box">Final Total: RM 0.00</div>

        <div id="notePOS" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 3: Stock Control (Increment & Decrement)</h2>
```

```
        <p>Simulating inventory updates.</p>

        Stock Level: <span id="stockVal" style="font-size: 1.5rem; font-
weight: bold;">10</span><br><br>

        <button onclick="runTask3('inc')">Add (+)</button>

        <button onclick="runTask3('dec')"
style="background:#6c757d;">Remove (-)</button>

        <div id="noteStock" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 4: Gate Access (Relational & Logical)</h2>

        <p>Rules: Age 18+ OR (Age 13+ AND Parent present).</p>

        Age: <input type="number" id="age" placeholder="Age">

        <label><input type="checkbox" id="parent"> With
Parent?</label><br><br>

        <button onclick="runTask4()" style="background:#27ae60;">Verify
Entry</button>

        <div id="outSecurity" class="output-box">Status: Waiting...</div>

        <div id="noteSecurity" class="short-note"></div>

    </section>
```

Inside the <script> tag, type all the following code:

JAVASCRIPT:

1.

```
<script>
    // Helper to show explanations
    function showNote(id, text) {
        const noteDiv = document.getElementById(id);
        noteDiv.style.display = 'block';
        noteDiv.innerHTML = text;
    }
```

2.

```
// --- TASK 1: ARITHMETIC (BASIC) ---
function runTask1() {
    console.log("--- Task 1: Arithmetic Operations ---");
    let students = 31;
    students = students + 1;   // Addition
    students = students - 2;   // Subtraction
    students = students * 2;   // Multiplication
    students = students / 2;   // Division
    let power = 10 ** 2;       // Exponentiation (100)
    let remainder = 31 % 2;    // Modulus (1 - Odd)

    console.log("Final Student Count: " + students);
    console.log("10 to the power of 2: " + power);
    console.log("Remainder of 31/2: " + remainder);
    alert("Check Browser Console (F12) for results!");
}
```

3.

```
// --- TASK 2: POS SYSTEM (ARITHMETIC & ASSIGNMENT) ---
function runTask2() {
    let price = 50;
    let qty = parseInt(document.getElementById('qty').value);

    // a. Arithmetic Operator (*)
    let total = price * qty;
    let note = "<b>Subtopic (a) Arithmetic:</b> <code>*</
    code> multiplied " + price + " by " + qty + ".<br>";

    // b. Assignment Operator (-=)
    if (qty > 4) {
        total -= 20; // Shorthand for total = total - 20
        note += "<b>Subtopic (b) Assignment:</b> <code>-=</
        code> updated the total by subtracting the discount.";
    }

    document.getElementById('outPOS').textContent = "Final
    Total: RM " + total.toFixed(2);
    showNote('notePOS', note);
}
```

4.

```
// --- TASK 3: INCREMENT & DECREMENT ---
let stock = 10;
function runTask3(type) {
    let note = "";
    if (type === 'inc') {
        stock++; // (c) Increment
        note = "<b>Subtopic (c) Increment:</b> <code>++</
        code> increased the stock by 1.";
    } else {
        stock--; // (c) Decrement
        note = "<b>Subtopic (c) Decrement:</b> <code>--</
        code> decreased the stock by 1.";
    }
    document.getElementById('stockVal').textContent = stock;
    showNote('noteStock', note);
}
```

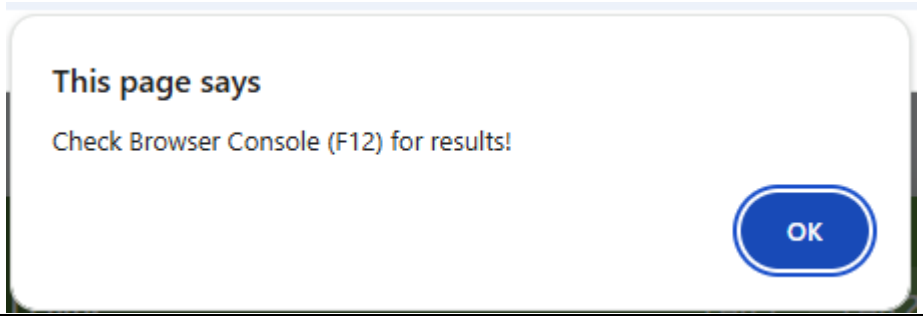
5.

```
    // --- TASK 4: RELATIONAL & LOGICAL ---
    function runTask4() {
        let age = parseInt(document.getElementById('age').value);
        let hasParent = document.getElementById('parent').checked;
        let status = document.getElementById('outSecurity');

        if (isNaN(age)) return alert("Please enter a valid age.");

        // d. Relational: >=
        // e. Logical: || (OR), && (AND)
        let isEligible = (age >= 18) || (age >= 13 && hasParent);

        status.textContent = isEligible ? "Status: ACCESS GRANTED
        ✅" : "Status: ACCESS DENIED ❌";
        status.style.color = isEligible ? "#27ae60" : "#ff4d4d";

        let note = "<b>Subtopic (d) Relational:</b> <code>>=</
        code> compared Age (" + age + ") to the limit.<br>";
        note += "<b>Subtopic (e) Logical:</b> Used <code>||</
        code> (OR) and <code>&&</code> (AND) to combine rules.";
        showNote('noteSecurity', note);
    }
</script>
```

Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

1.

Screenshot of the output



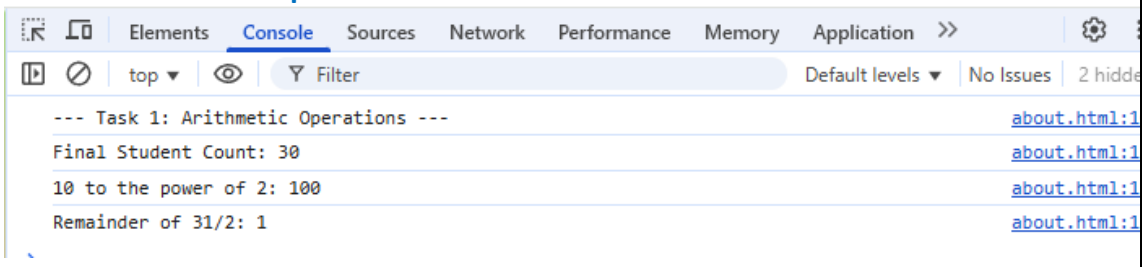
Explanation

1. Browser Alert Box

An **alert()** function is used to trigger a pop-up window. This serves as a notification to the user, instructing them to open the Developer Tools (F12) to view the actual data output in the console.

2.

Screenshot of the output

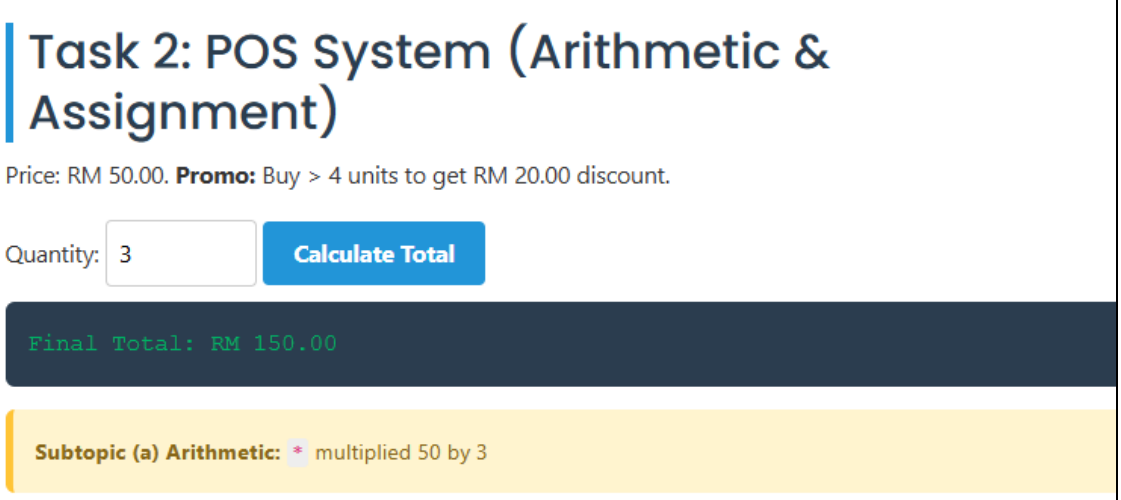


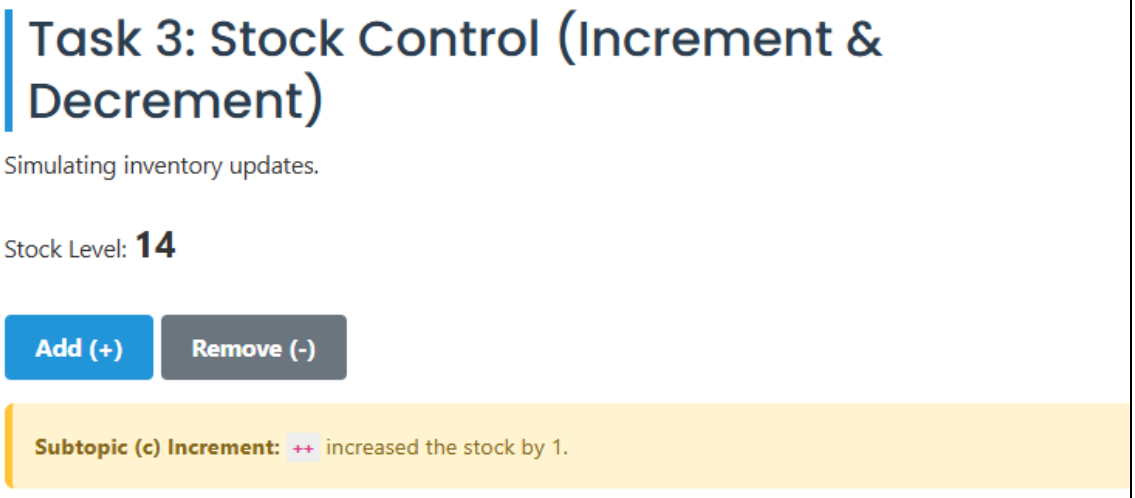
Explanation

2. Console Output (Arithmetic Operations)

The JavaScript **console.log()** method is used to display the results of mathematical operations. It shows:

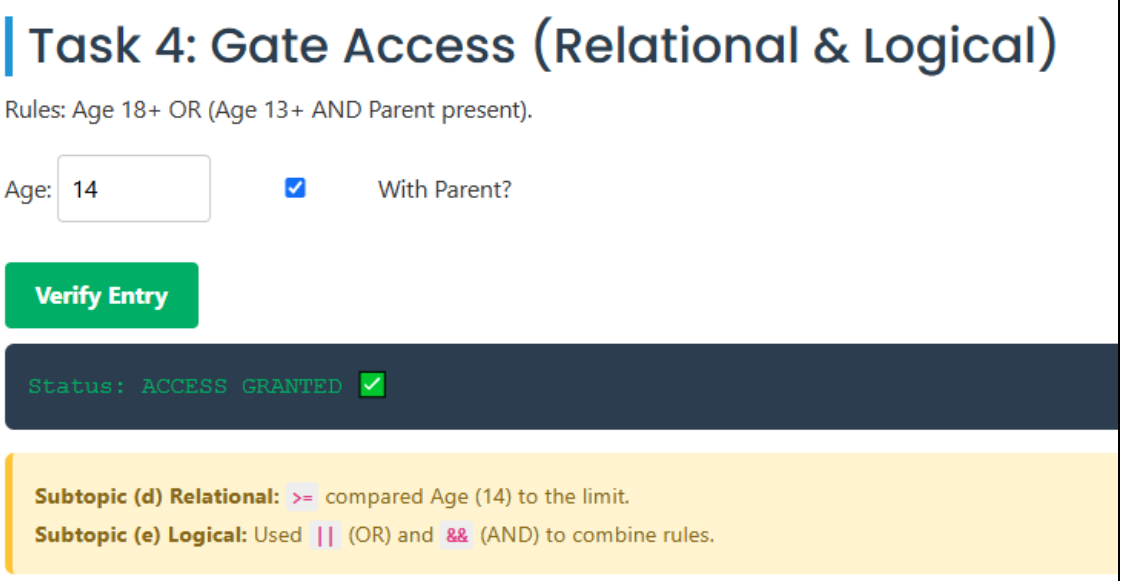
- **Addition:** Combining student counts.
- **Exponentiation:** Calculating $10^2 = 100$.
- **Modulus:** Finding the remainder of $31 \div 2$, which is 1.

3.	**Screenshot of the output** 
	Explanation 3. POS System Logic This demonstrates **interactive arithmetic and assignment**. When the "Calculate Total" button is clicked, a script takes the quantity input (3) and multiplies it by the unit price (RM 50.00) to assign the final total (RM 150.00) to the display area.

4.	**Screenshot of the output** 
	Explanation 4. Stock Control (Increment & Decrement) This task demonstrates **unary operators** used to update inventory levels:

- **Increment (++):** Clicking "Add (+)" increases the stock level by 1.
- **Decrement (--):** Clicking "Remove (-)" decreases the stock level by 1.
- This simulates real-time data updates within an application.

5. **Screenshot of the output**



Explanation

5. Gate Access (Relational & Logical)

This task uses complex conditions to determine entry permissions:

- **Relational Operators (>=):** These check if the input age meets specific limits (e.g., age 14 compared to the 13+ limit).
- **Logical Operators (`||` OR, `&&` AND):** These combine multiple rules to create a final "True" or "False" result.
- **Result:** In the example, access is granted because the person is over 13 **and** a parent is present.

3. Quick Check (Reflection)

Instruction: Answer the following questions based on your coding results.

1. **Modulus Utility:** Based on Task 1, if students % 2 returns 0, is the number even or odd?

   = If `students % 2` returns **0**, the number is **even**.

2. **Shorthand Power:** In Task 2, rewrite total -= 20 using the standard assignment method.

   = total = total - 20

3. **Logical Priority:** In Task 4, why did we put (age >= 13 && hasParent) in parentheses?

   = In Task 4, the parentheses were used around `(age >= 13 && hasParent)` to **ensure correct operator precedence**.

4. **Time Calculation:** (Extra Challenge) If a variable totalMinutes = 130, write the formula to find the remaining minutes after converting to hours using the **Modulus** operator.

   = remainingMinutes = totalMinutes % 60;